

Sim-to-Real Transfer Limitations in Reinforcement Learning for Partially Observable Mars Rover Navigation: A Case Study Using Real Jezero Crater Terrain

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Abstract

Reinforcement learning (RL) agents for planetary rover navigation are typically trained and evaluated in synthetic simulation environments, under the implicit assumption that synthetic performance predicts real-world capability. This study tests that assumption directly. Deep Q-Network (DQN) and Proximal Policy Optimization (PPO) agents were trained on a custom partially observable navigation task, in which an agent observes only a local 13×13 terrain patch rather than the full map, across three synthetically generated difficulty tiers, with each configuration trained across five independent random seeds. Both algorithms achieved strong synthetic performance (mean episodic reward 91–114 on easy and medium terrain). The same, unmodified trained agents were then evaluated on real digital elevation data from Jezero Crater, Mars, the Perseverance rover's active landing site, using terrain classifications calibrated against Perseverance's actual driven path. Multi-seed real-terrain success rates were substantially lower than synthetic performance predicted ($2.0 \pm 1.4\%$ to $37.2 \pm 8.8\%$ across models under deterministic action selection), revealing a severe sim-to-real gap. Systematic analysis identified the dominant failure mode as a deterministic policy cycle: agents became trapped in short, exactly repeating position loops, accounting for 100% of all timeout failures across every model tested under deterministic action selection. This behavior was substantially more prevalent in PPO than in DQN, and switching PPO to stochastic action selection reduced its cycle rate to near zero and increased its real-terrain success rate by 12–40 percentage points across difficulty tiers, while the same intervention did not meaningfully benefit DQN. SHAP-based feature attribution analysis confirmed algorithm-specific mechanistic signatures: PPO cycling episodes showed near-parity between direction-to-goal and terrain-patch signals, while DQN failures were characterized by systematically reduced terrain-patch attention. These findings indicate that synthetic benchmark performance is an unreliable predictor of real-world navigation capability under partial observability, and that the reliability of this predictor differs systematically by algorithm family.

Keywords: reinforcement learning, sim-to-real transfer, partial observability, Mars rover navigation, DQN, PPO, multi-seed evaluation

Introduction

Autonomous navigation is a central capability requirement for planetary rovers, which must traverse terrain using only onboard sensing rather than a complete map of their surroundings (Kaelbling et al., 1998). The Perseverance rover, for instance, relies on its AutoNav system to autonomously evaluate terrain and plan safe paths, having covered 88% of its first Mars year of driving without direct human review (Verma et al., 2023). Real planetary rovers such as Perseverance navigate using onboard stereo vision and local terrain assessment, with

AutoNav operating over a sensing range of roughly 6 meters ahead of the rover (Verma et al., 2023). This constraint places rover navigation squarely within the class of partially observable Markov decision processes (POMDPs): the true environmental state, the full terrain map, is never directly available to the agent, which must instead act on a limited local observation. Reinforcement learning (RL) offers a natural framework for learning navigation policies under this constraint, and deep RL methods such as Deep Q-Networks (DQN; Mnih et al., 2015) and Proximal Policy Optimization (PPO; Schulman et al., 2017) have demonstrated strong performance on a range of sequential decision-making tasks.

A methodological challenge specific to this domain, however, is rarely addressed explicitly: RL agents intended for real-world deployment are almost always trained and evaluated in synthetic simulation, and the assumption that synthetic performance transfers to the real environment is frequently left implicit rather than tested (Zhao et al., 2020). This is a nontrivial assumption, and one that the sim-to-real transfer literature has documented extensively in other robotic domains (Zhao et al., 2020). Synthetic terrain generation procedures typically differ from real terrain in systematic ways (for example, in how obstacles are spatially distributed), and any such difference has the potential to invalidate conclusions drawn purely from simulation.

This study investigates two related but logically distinct questions. First, can an RL agent learn effective navigation using only a local, partial observation of its surroundings? Second, and more fundamentally, is performance measured in a synthetic training environment a trustworthy indicator of real-world capability at all? We argue, and provide direct empirical evidence, that the second question is a prerequisite for the first to be answerable: without validating the training environment against real data, an apparently successful navigation policy may in fact be exploiting properties of the synthetic environment that do not hold.

To address these questions, we developed a custom Gymnasium-compatible (Towers et al., 2024) partially observable navigation environment, trained DQN and PPO agents using Stable-Baselines3 (Raffin et al., 2021) on synthetically generated terrain across five independent random seeds, and subsequently evaluated the trained agents, without any retraining or fine-tuning, on real terrain derived from Digital Elevation Model (DEM) data of Jezero Crater, the landing site of NASA's Perseverance rover. Multi-seed evaluation allows us to report mean $\pm$ standard deviation performance estimates, addressing a methodological limitation common in prior single-seed RL studies. We report a substantial sim-to-real performance gap, identify its dominant mechanism, and evaluate candidate mitigations including stochastic action selection and SHAP-based feature attribution.

Method

Environment Design

A custom environment, MarsTerrainEnv, was implemented using the Gymnasium API (Towers et al., 2024). The environment represents terrain as a 50×50 grid of four cell types: safe ground, rock (passable but rougher), boulder (impassable), and crater (impassable). The agent occupies a single grid cell and may move in one of eight compass directions per timestep.

Critically, the agent's observation is not the full grid. At each timestep the agent receives only a 13×13 local patch of terrain centered on its current position, together with a normalized

direction vector and scalar distance to the goal. The 13x13 window size (representing approximately 6 meters of terrain at the 0.5 m/cell grid resolution) is motivated by the sensing range of Perseverance's AutoNav system (Verma et al., 2023). The full map, including the location and extent of any obstacles outside this local window, is never exposed to the agent. This design instantiates the partial-observability constraint central to the study's research questions.

The reward function was defined as follows: +100 for reaching the goal cell, -10 for a collision (attempting to move into impassable terrain or off the grid boundary, which also ends the episode), +0.5 for a move that decreased Euclidean distance to the goal, -1 for a move that increased it, and a constant -0.1 applied to every step to encourage efficient paths. Episodes were capped at 200 steps, after which an unresolved episode was recorded as a timeout.

Synthetic terrain was generated procedurally by sampling each grid cell independently from a difficulty-specific categorical distribution over the four terrain types, with three difficulty tiers (easy, medium, hard) differing in obstacle density. Every generated map was verified, via breadth-first search, to contain a traversable path between the randomly placed start and goal positions before being used for training or evaluation.

Reinforcement Learning Algorithms

Two algorithms were trained on the synthetic environment using Stable-Baselines3 (Raffin et al., 2021): DQN (Mnih et al., 2015), an off-policy, value-based method, and PPO (Schulman et al., 2017), an on-policy, policy-gradient method trained with an entropy bonus to encourage exploration. Both used a MultiInputPolicy architecture with a small convolutional branch for the terrain patch concatenated with a multilayer-perceptron branch for the direction and distance features. Each algorithm was trained independently on each of the three synthetic difficulty tiers for 500,000 timesteps.

An initial DQN training run on medium-difficulty terrain exhibited clear signs of value-function instability (a persistently increasing training loss); doubling the training budget to 1,000,000 timesteps did not resolve this. Reducing the learning rate (1×10^{-4} to 5×10^{-5}) and increasing target-network update frequency resolved the instability; this adjusted configuration was used for all DQN training reported here.

To assess variance across training runs, a methodological limitation of single-seed RL studies, all six model configurations (DQN and PPO $\times$ three difficulty tiers) were trained with five independent random seeds (seeds 0–4), yielding 30 total training runs. Models already trained were cached and skipped on subsequent runs, allowing incremental seed extension without redundant computation.

Real-World Terrain Acquisition and Validation

Real terrain data were obtained from a Context Camera (CTX) Digital Elevation Model mosaic of Jezero Crater (20 m/pixel resolution), distributed by the United States Geological Survey (USGS) Astrogeology Science Center. Terrain was classified by computing local slope from the elevation gradient and applying slope thresholds to assign each cell to the same four categories used in synthetic terrain generation.

Slope thresholds were calibrated against the actual recorded driving path of the Perseverance rover, obtained from NASA's Mars 2020 PLACES data product. An initial

calibration attempt classified 32.7% of real-driven terrain as impassable, indicating miscalibration. Thresholds were re-derived empirically from the slope distribution traversed by the rover (the 50th, 95th, and 99th percentiles, yielding thresholds of 4.5°, 15.5°, and 18.5°), reducing the mismatch to 4.9%.

Evaluation Procedure

Real-terrain maps were sampled from the DEM using a stratified procedure: the DEM was tiled, mean slope was computed per tile, and tiles were binned into five difficulty strata. Ten solvable start/goal configurations were generated per stratum (50 maps total), each verified for reachability. Each trained model was evaluated on the full 50-map real-terrain set under both deterministic (greedy) and stochastic (sampled) action-selection modes. Across five seeds, each configuration produced 250 evaluation episodes per action-selection mode.

Performance metrics (success rate, collision rate, timeout rate, cycle rate, and mean episodic reward) were computed per seed and aggregated as mean $\pm$ standard deviation across seeds.

Feature Attribution (SHAP) Analysis

SHAP (SHapley Additive exPlanations; Lundberg & Lee, 2017) was used to decompose each policy's action choice into the contribution of three grouped observation components: the local terrain patch, the direction-to-goal vector, and the distance-to-goal scalar. The three components were treated as a 3-player coalitional game, with a missing component replaced by its mean value across a background sample of real-terrain states. Shapley values were computed exactly (all $2^3 = 8$ coalitions enumerated via `shap.KernelExplainer`). SHAP analysis was conducted for DQN and PPO at medium and hard difficulty (seed 0), under both deterministic and stochastic action selection, yielding eight analysis runs total.

Results

Synthetic Terrain Training

Both DQN and PPO achieved strong performance on synthetic training terrain (Table 1). Performance was comparable between algorithms on easy and medium difficulty. On hard difficulty, DQN's evaluation reward remained negative even after doubling the training budget, while PPO achieved substantially higher reward (26.42). As reported below, this synthetic-terrain ranking did not hold under real-terrain evaluation.

Table 1

Synthetic Terrain Evaluation Performance by Algorithm and Difficulty

Algorithm	Difficulty	Mean Reward	Mean Episode Length
DQN	Easy	97.87	37.1
DQN	Medium	113.19	34.2
DQN	Hard (500k steps)	-14.76	158.1
DQN	Hard (1M steps)	-16.64	150.2
PPO	Easy	111.81	32.2
PPO	Medium	103.60	41.8
PPO	Hard	26.42	97.4

Real-Terrain Evaluation (Multi-Seed)

Table 2 reports real-terrain performance across five seeds under deterministic action selection. Real-terrain success rates were substantially and consistently lower than synthetic evaluation performance suggested, across every model. Unlike the single-seed results reported previously, multi-seed evaluation reveals that no single model configuration consistently dominated: DQN Easy ($32.0 \pm 8.0\%$) and PPO Easy ($37.2 \pm 8.8\%$) achieved the highest mean success rates under deterministic selection, while DQN Hard remained catastrophically low ($2.0 \pm 1.4\%$). The cycle rate equaled the timeout rate in all cases, confirming that 100% of timeout episodes were attributable to deterministic policy cycles across all seeds.

Table 2

Real-Terrain Evaluation Performance (50 Stratified Maps $\times$ 5 Seeds, Deterministic Action Selection, Mean $\pm$ SD)

Model	Success	Collision	Timeout	Cycle	Mean Reward
DQN Easy	$32.0 \pm 8.0\%$	$59.2 \pm 7.4\%$	$8.8 \pm 8.1\%$	$8.8 \pm 8.1\%$	29.84 ± 11.33
DQN Medium	$30.8 \pm 9.5\%$	$36.8 \pm 18.0\%$	$32.4 \pm 19.4\%$	$32.4 \pm 19.4\%$	16.03 ± 20.02
DQN Hard	$2.0 \pm 1.4\%$	$3.6 \pm 1.5\%$	$94.4 \pm 1.7\%$	$94.4 \pm 1.7\%$	-57.96 ± 2.75
PPO Easy	$37.2 \pm 8.8\%$	$25.2 \pm 5.4\%$	$37.6 \pm 10.7\%$	$37.6 \pm 10.7\%$	12.57 ± 18.08
PPO Medium	$28.8 \pm 4.6\%$	$14.8 \pm 3.8\%$	$56.4 \pm 6.2\%$	$56.4 \pm 6.2\%$	-8.18 ± 10.29
PPO Hard	$25.6 \pm 7.4\%$	$6.4 \pm 3.0\%$	$68.0 \pm 9.2\%$	$68.0 \pm 9.2\%$	-18.01 ± 14.81

Note. Collision rate is computed as $100\% - \text{Success}\% - \text{Timeout}\%$. Cycle rate equals timeout rate in all cases (100% of timeouts were policy cycles). SD = standard deviation across 5 seeds.

Failure Mode: Deterministic Policy Cycling

Manual inspection of individual failure trajectories revealed that agents frequently became trapped in short, exactly repeating position sequences rather than exhibiting open-ended confusion. For example, one PPO trajectory on a real-terrain map repeated the identical three-position sequence $(1,7) \rightarrow (1,6) \rightarrow (2,7)$ for the remainder of a 200-step episode. Systematic cycle detection across the full multi-seed evaluation set confirmed that this was not an isolated case: across all five seeds and all six model configurations, 100% of timeout episodes under deterministic action selection were attributable to a detected short-period cycle (period ≤ 6 , repeating ≥ 3 times). This finding was entirely consistent across seeds: the cycle rate standard deviation matched the timeout rate standard deviation in every case.

This behavior is consistent with the well-documented phenomenon of perceptual aliasing in partially observable domains (Chrisman, 1992): under a deterministic policy, if the agent returns to a position that produces an identical observation to one previously encountered, it will necessarily select the same action. Absent any memory of prior states, such a recurrence is guaranteed to repeat indefinitely.

Obstacle Geometry Analysis

To test whether obstacle geometry contributes to the observed sim-to-real gap, connected-component cluster statistics were computed for 150 freshly generated synthetic maps (50 per difficulty tier) and the 50 real-terrain evaluation maps (Table 3). Real-terrain obstacle clusters were substantially larger on average than synthetic clusters (mean 15.90 cells versus 2.43 cells, a 6.6-fold difference), and the maximum observed cluster size differed even more sharply (1,282 cells in real terrain versus 79 in synthetic terrain, on a 2,500-cell grid). Mean

elongation was similar between terrain types (1.39 real versus 1.24 synthetic), indicating the primary geometric difference is one of scale rather than shape.

Table 3

Obstacle Cluster Geometry: Synthetic Versus Real Terrain

Metric	Synthetic Terrain	Real Terrain
Mean cluster size (cells)	2.43	15.90
Median cluster size (cells)	1.0	2.0
Maximum cluster size (cells)	79	1,282
Mean elongation	1.24	1.39
Single-cell clusters	57.1%	48.5%

Effect of Stochastic Action Selection

Because the cycling failure mode depends mechanistically on deterministic action selection, all models were re-evaluated on the 50-map real-terrain set using stochastic action selection. Results diverged sharply by algorithm (Table 4). For PPO, stochastic action selection produced large and consistent improvements across all three difficulty tiers, with success rate increases of 12–40 percentage points and near-complete elimination of cycling behavior (cycle rate dropped to $0.0 \pm 0.0\%$ on medium and hard). For DQN, the same intervention produced no meaningful benefit and, on easy and medium terrain, a small decrease in success rate.

Table 4

Effect of Stochastic Action Selection on Real-Terrain Success Rate (Mean $\pm$ SD, 5 Seeds)

Model	Det. Success	Stoch. Success	Change	Det. Cycle	Stoch. Cycle
DQN Easy	$32.0 \pm 8.0\%$	$30.4 \pm 7.0\%$	−1.6 pts	$8.8 \pm 8.1\%$	$0.4 \pm 0.9\%$
DQN Medium	$30.8 \pm 9.5\%$	$37.6 \pm 9.3\%$	+6.8 pts	$32.4 \pm 19.4\%$	$7.6 \pm 11.5\%$
DQN Hard	$2.0 \pm 1.4\%$	$5.2 \pm 1.1\%$	+3.2 pts	$94.4 \pm 1.7\%$	$50.0 \pm 2.8\%$
PPO Easy	$37.2 \pm 8.8\%$	$49.6 \pm 3.8\%$	+12.4 pts	$37.6 \pm 10.7\%$	$0.0 \pm 0.0\%$
PPO Medium	$28.8 \pm 4.6\%$	$52.8 \pm 7.9\%$	+24.0 pts	$56.4 \pm 6.2\%$	$0.0 \pm 0.0\%$
PPO Hard	$25.6 \pm 7.4\%$	$65.6 \pm 5.7\%$	+40.0 pts	$68.0 \pm 9.2\%$	$0.0 \pm 0.0\%$

This asymmetry is consistent with a specific mechanistic account. PPO is trained with an explicit entropy bonus that discourages the policy from collapsing to a single dominant action, meaning the greedy and near-greedy actions are typically close in probability; introducing sampling therefore costs little in expected quality while reliably breaking exact repetition. DQN estimates a single scalar value per action with no comparable pressure toward closely spaced alternatives, so a sampled non-greedy action is more likely to be a genuinely worse choice.

Feature Attribution (SHAP) Analysis

SHAP-based feature-group attribution was computed for DQN and PPO at medium and hard difficulty (seed 0), under both deterministic and stochastic action selection (Table 5). Two distinct, algorithm-specific patterns emerged that replicate and extend the original single-seed findings.

For PPO, the ratio of direction-to-goal attribution to terrain-patch attribution was consistently higher during successful episodes than during cycling episodes under deterministic selection: 1.20 versus 0.61 on medium terrain, and 1.62 versus 0.73 on hard terrain. In every

case, cycling episodes showed the two signals close to parity (ratio near 1.0), while successful episodes showed direction-to-goal clearly dominant. Under stochastic action selection, cycling episodes disappear almost entirely for PPO, and the surviving success episodes continue to show direction dominance, confirming that the near-parity condition is specifically associated with the cycling failure mode, not with PPO's policy structure in general.

For DQN, the pattern instead involved the absolute magnitude of terrain-patch attribution rather than its relative balance with direction-to-goal. Under deterministic selection, patch attribution was lower during failure outcomes than during success across medium terrain (cycle: 1.268, success: 1.162) while direction-to-goal attribution remained comparatively stable. DQN Hard showed patch attribution completely dominating during cycling (5.41), indicating the policy had collapsed to almost exclusively local terrain-reactive behavior. Under stochastic selection, DQN Hard produced a novel "timeout" outcome category (distinct from cycle) with even higher patch attribution (6.62), suggesting that randomized action selection produces a different failure mode (prolonged wandering driven by terrain-patch signals) rather than eliminating failure altogether.

Table 5

Mean |Shapley Value| by Feature Group and Outcome: DQN and PPO, Medium and Hard (Seed 0)

Model	Mode	Outcome	N	Patch	Direction	Distance	Dir/Patch
DQN Medium	Det.	Collision	23	0.866	0.759	2.323	0.88
DQN Medium	Det.	Cycle	103	1.268	0.662	2.387	0.52
DQN Medium	Det.	Success	24	1.162	0.903	2.531	0.78
DQN Medium	Stoch.	Collision	60	1.283	0.753	2.838	0.59
DQN Medium	Stoch.	Cycle	27	1.598	0.564	0.990	0.35
DQN Medium	Stoch.	Success	63	0.951	0.859	2.457	0.90
DQN Hard	Det.	Cycle	149	5.412	1.105	1.290	0.20
DQN Hard	Det.	Success	1 (n.r.)	11.827	0.787	0.083	n/a
DQN Hard	Stoch.	Collision	23	3.360	1.186	1.138	0.35
DQN Hard	Stoch.	Cycle	87	5.049	0.998	1.392	0.20
DQN Hard	Stoch.	Timeout	40	6.616	1.197	1.404	0.18
PPO Medium	Det.	Cycle	134	0.371	0.227	0.007	0.61
PPO Medium	Det.	Success	13	0.279	0.335	0.004	1.20
PPO Medium	Stoch.	Collision	78	0.243	0.301	0.008	1.24
PPO Medium	Stoch.	Success	72	0.180	0.300	0.009	1.67
PPO Hard	Det.	Cycle	145	0.327	0.239	0.004	0.73
PPO Hard	Det.	Success	5	0.217	0.353	0.004	1.62
PPO Hard	Stoch.	Collision	35	0.287	0.304	0.003	1.06
PPO Hard	Stoch.	Success	102	0.284	0.288	0.002	1.01
PPO Hard	Stoch.	Timeout	13	0.290	0.165	0.004	0.57

Note. n.r. = not a reliable estimate given sample size. SHAP magnitudes are not comparable across DQN and PPO due to difference in scale between Q-values and action probabilities. Det. = deterministic action selection; Stoch. = stochastic action selection.

Discussion

Synthetic Performance Does Not Predict Real-World Capability

The central finding of this study is that strong performance in a synthetic training environment did not predict real-world performance, and in one case (DQN versus PPO on hard-difficulty terrain in the original single-seed evaluation) produced a conclusion that was reversed under real-terrain evaluation. Multi-seed evaluation reinforces and extends this finding: the high variance observed across seeds (e.g., DQN Medium success rate: $30.8 \pm 9.5\%$) demonstrates that even a single real-terrain evaluation run can be misleading. The original single-seed result of 60% for DQN Medium fell more than two standard deviations above the multi-seed mean, illustrating precisely the risk that motivates multi-seed reporting.

This finding should not be read as evidence that the reinforcement learning approach is ineffective. Rather, it identifies a specific, mechanistically understood limitation, namely deterministic policy cycling under perceptual aliasing, that was invisible under synthetic evaluation because the synthetic terrain generation procedure rarely produces the large, contiguous obstacle structures that expose this limitation. Real terrain, arising from spatially continuous elevation data, produces such structures routinely.

A Testable, Confirmed Mechanistic Account

The cycling failure mode is well explained by perceptual aliasing (Chrisman, 1992) interacting with a reward function that shapes behavior primarily via straight-line distance to the goal, which is functionally equivalent to an artificial potential field with documented susceptibility to local minima near large obstacles (Koren & Borenstein, 1991). The multi-seed results confirm that this account is not a single-seed artifact: 100% of timeout episodes were policy cycles across all five seeds and all six model configurations.

The SHAP analysis provides direct, model-internal evidence for algorithm-specific mechanistic differences. For PPO, cycling episodes consistently showed near-parity between direction-to-goal and terrain-patch signals (ratios 0.61–0.73), while successful episodes showed direction dominance (1.20–1.62). For DQN, failures were instead characterized by high absolute patch attribution in hard terrain (5.41 during cycles) with direction attribution suppressed, suggesting a qualitatively different failure mode: the policy becomes excessively reactive to local terrain features while neglecting goal-directed navigation. The stochastic SHAP results extend this picture: for PPO, removing cycles through stochastic selection reveals that the remaining (success and collision) episodes show healthy direction dominance, confirming the cycling condition was specifically the near-parity state. For DQN Hard, stochastic selection introduces a distinct timeout category with even higher patch attribution (6.62), consistent with a policy that wanders reactively to terrain without goal-directed resolution.

Limitations

Several limitations qualify these findings. First, while multi-seed evaluation substantially strengthens the reported results, five seeds per configuration remains a modest sample; the standard deviations reported (particularly for DQN Medium: ± 9.5 – 19.4%) indicate residual variance that a larger seed count would further characterize. Second, the stratified real-terrain evaluation set remains limited to 50 maps drawn from one DEM tile covering one region of Jezero Crater; broader geographic sampling would strengthen generality. Third, the SHAP analysis was conducted for seed 0 only; extending it across seeds would confirm whether the mechanistic patterns are themselves stable across training runs. Fourth, terrain classification thresholds remain a modeling simplification: real traversability depends on additional factors not captured by slope alone.

Implications and Future Work

These results suggest three practical recommendations for RL-based rover navigation research. First, multi-seed evaluation should be treated as a minimum standard, not an optional refinement. Single-seed real-terrain results can deviate substantially from the true mean, as demonstrated by the DQN Medium case. Second, real-terrain or real-data validation should be treated as a required component of evaluation, given the demonstrated capacity for synthetic-only evaluation to produce misleading or reversed rankings. Third, because the identified failure mode is attributable in part to a lack of agent memory, architectures that retain information about recently visited states (recurrent policies, explicit visitation history in the observation) represent a natural next step, alongside investigation of domain randomization over obstacle cluster-size distributions to close the geometric sim-to-real gap identified here.

Conclusion

This study trained and validated reinforcement learning agents for partially observable Mars rover navigation using both synthetic terrain and real, ground-truth-calibrated terrain from Jezero Crater, across five independent random seeds. While synthetic evaluation suggested strong and reliable navigation capability for both DQN and PPO, real-terrain multi-seed evaluation revealed a substantial sim-to-real gap (success rates of $2.0 \pm 1.4\%$ to $37.2 \pm 8.8\%$ under deterministic action selection), driven predominantly by a deterministic policy-cycling failure mode affecting 100% of timeout episodes across all seeds and model configurations. This failure mode was largely mitigable for PPO through stochastic action selection (+12 to +40 percentage points across difficulty tiers, with cycle rates dropping to $0.0 \pm 0.0\%$), but not for DQN. SHAP-based feature attribution confirmed algorithm-specific mechanistic signatures that are consistent across both deterministic and stochastic evaluation modes. Taken together, these results demonstrate that environment validation against real-world data and multi-seed reporting are necessary conditions for RL-based navigation research conclusions to be trustworthy.

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